EMPLOYEE ATTRITION ANALYSIS

Problem

A company is facing **high employee attrition** but doesn't know:

1. Why employees are leaving
2. Which departments are most affected
3. What factors (salary, workload, satisfaction) drive attrition

Goal

Identify key drivers of attrition and suggest actions to reduce it

Questions to answer

1. Job involvement vs attrition
2. Male vs female attrition
3. Attrition age group wise
4. Month wise attrition
5. Dept wise attrition
6. Job role wise attrition
7. Attrition based on experience group
8. Year since last promotion vs attrition
9. Monthly income group vs attrition
10. Salary hike group vs attrition
11. Overtime vs attrition
12. Environment Satisfaction vs attrition
13. Job Satisfaction vs attrition
14. Work Life Balance vs attrition

Analysis Using SQL

1. **Overall attrition rate**

```
select (
        select count(*) from learning_projects.`hr dataset`where Attrition_flag=1
)*100/count(attrition_flag) as attrition_percent
from learning_projects.`hr dataset`
```

	attrition_percent
▶	16.1224

2. **Attrition percent dept wise**

```
select department,round(sum(attrition_flag)*100/count(*),2) as attrition_percent from
learning_projects.`hr dataset`
group by department
order by attrition_percent desc
```

department	attrition_percent
Sales	20.63
Human Resources	19.05
Research & Development	13.84

3. **For highest attrition dept find attrition based on overtime**

```
select Over_Time, sum(attrition_flag)*100/count(*) as attrition_percent
from learning_projects.`hr dataset`
where department = 'Sales'
group by Over_Time
```

Over_Time	attrition_percent
1	37.5000
0	13.8365

4. **For highest attrition dept find attrition based on Monthly income group**

```
select Monthly_Income_Group, sum(attrition_flag)*100/count(*) as attrition_percent
from learning_projects.`hr dataset`
where department = 'Sales'
group by Monthly_Income_Group
```

Monthly_Income_Group	attrition_percent
Medium	15.3005
High	17.2043
Low	41.5584

5. **Attrition vs combination of Monthly income group and over time.**

```
select Monthly_Income_Group,Over_Time, sum(attrition_flag)*100/count(*) as attrition_percent
from learning_projects.`hr dataset`
where department = 'Sales'
group by Monthly_Income_Group, Over_Time
```

Monthly_Income_Group	Over_Time	attrition_percent
Medium	0	8.3969
High	0	12.7820
High	1	28.3019
Low	0	29.6296
Medium	1	32.6923
Low	1	69.5652

6. **Job satisfaction vs attrition**

```
select Job_Satisfaction, sum(attrition_flag)*100/count(*) as attrition_percent
from learning_projects.`hr dataset`
-- where department = 'Sales'
group by Job_Satisfaction
```

Job_Satisfaction	attrition_percent
1	22.8374
3	16.5158
2	16.4286
4	11.3290

INSIGHTS

1. **Overall Attrition**
    a. Overall attrition rate is **~16%**
    b. Indicates moderate employee churn requiring targeted action

2. **Department-Level Insight**
    a. Sales department has highest attrition
    b. Followed by Human Resources
    c. Suggests role-specific pressure or compensation issues

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5. **Compensation Impact**
    a. **Low salary group shows highest attrition (~26%)**
    b. Medium and High salary groups are significantly lower (~11–12%)

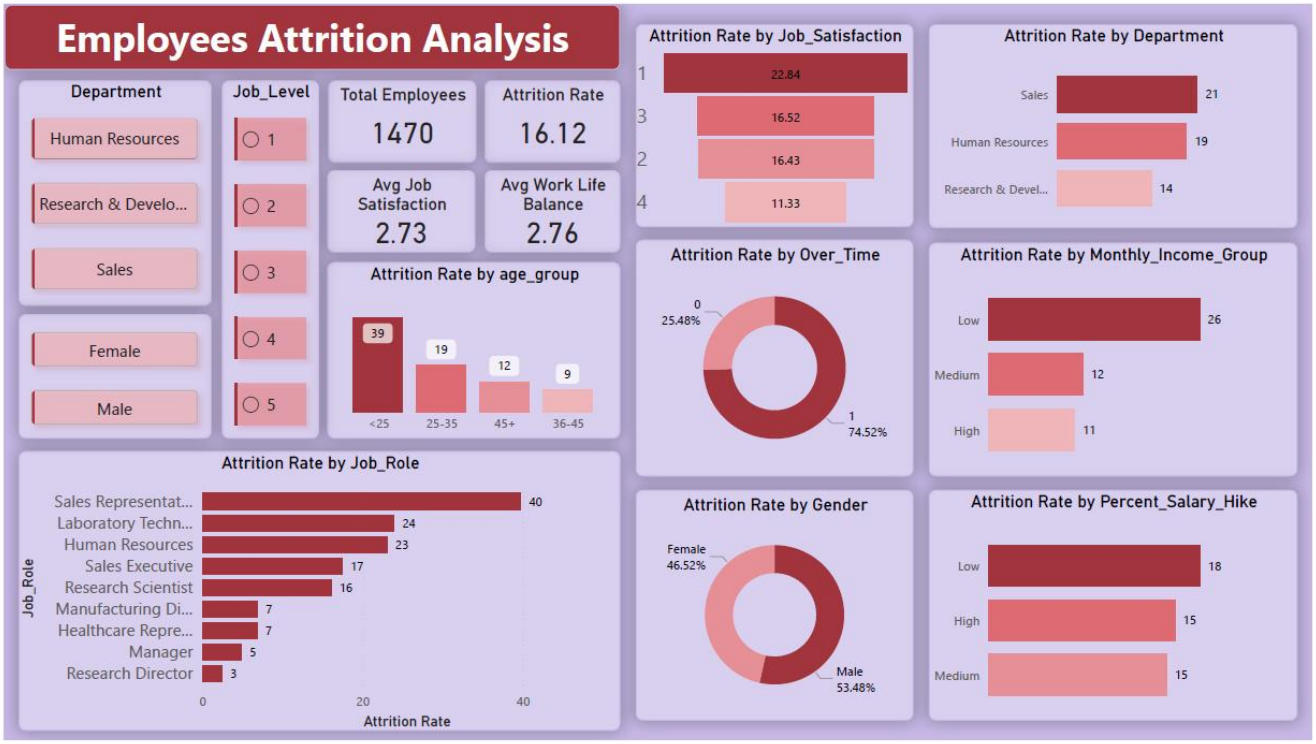
6. **Overtime Impact**
    a. Employees working overtime show **~74% attrition**
    b. Non-overtime employees show much lower attrition (~25%)

7. **Job Role Analysis**
    a. **Sales Representatives** and **Lab Technicians** show highest attrition
    b. Indicates operational or role-specific stress

8. **Demographics**
    a. Gender difference is minimal → not a key driver
    b. Younger employees (<25) show higher attrition
    c. Suggests early-career instability

9. **Salary Hike Impact**
    a. Lower salary hike group shows slightly higher attrition
    b. Impact exists but weaker than base salary

Dashboard



Data Cleaning using Python

Code:

```
pip install pandas

import pandas as pd

df = pd.read_csv('hr dataset.csv')

df.info()

df['Department'].unique()

pd.set_option('display.max_columns', None)

df.head()

df['Over_Time'].unique()

df['Over_Time'] = df['Over_Time'].map({'Yes':1,'No':0})

df['Over_Time'].unique()

df['Percent_Salary_Hike'] = pd.qcut(df['Percent_Salary_Hike'], q=3, labels=['Low', 'Medium', 'High'])

df['Monthly_Income_Group'] = pd.qcut(df['Monthly_Income'], q=3, labels=['Low', 'Medium', 'High'])

df['Monthly_Income_Group'].unique()
```

```
df.isnull().sum()

from sqlalchemy import create_engine

engine = create_engine("mysql+pymysql://root:root%40123@localhost:3306/learning_projects")

df.to_sql(
    name='hr_dataset',
    con=engine,
    index=False,
    if_exists='replace'
)
```

Summary

The analysis reveals that employee attrition is primarily driven by compensation and workload factors. The Sales department experiences the highest attrition, with Sales Representatives being the most affected role. Employees in the low salary group and those working overtime are significantly more likely to leave, with the combined segment showing nearly 70% attrition, making it the most critical risk group. Additionally, lower job satisfaction contributes to higher attrition, while demographic factors such as gender show minimal impact. Overall, the findings suggest that improving compensation structures, reducing excessive overtime, and enhancing employee satisfaction can significantly reduce attrition.

Recommendations

1. Review and improve compensation for low salary employees
2. Reduce overtime or redistribute workload
3. Focus retention strategies in Sales department
4. Improve job satisfaction through engagement programs
5. Monitor high-risk groups (low salary + overtime)